MLOps demonstration evidence

Report assembled 07 Nov 2026, 09:08 IST. Evidence timestamps below identify when each check ran.

Evidence	Recorded result	When checked
Current Python suite	58/58 passed; 0 failures; 0 errors; 0 skipped	07 Nov 2026, 08:51 IST
Local runtime	overall_passed=True; 8 checks	07 Nov 2026, 08:51 IST
MLflow registry	v1 READY; approval match=True	07 Nov 2026, 08:53 IST
GitHub CI	completed / success	07 Nov 2026, 08:54 IST
Historical release quality	quality_gate_passed=True	06 Nov 2026, 12:25 IST

Tools and their role

Pytest checks the pipeline, API and recovery paths. pandas and scikit-learn support historical quality evaluation. MLflow tracks experiments and registers the versioned model; Joblib stores its artifact. FastAPI/Pydantic/Uvicorn serve inference. Docker Desktop, WSL 2 and Compose run the local service. GitHub Actions tests/builds the image; GHCR stores the published release.

Deployment boundary: GitHub Actions automates tests, image build, publication and published-image verification. A local command starts the scripted workstation deployment. This report does not claim automatic deployment from GitHub to the Windows PC.

Model quality and approved scope

Gradient Boosting Regressor | cold_start_v1 | historical approval recorded 06 Nov 2026, 12:25 IST

Approved use: Route 90, eastbound segment ORR_CHIRAG_TO_NEHRU_PLACE_EB. Target: segment riding time in minutes.

Split	IST observation dates	Traversals
Train	2019-07-01, 2019-07-02, 2019-07-03	408
Validation	2019-07-05	102
Test	2019-07-08	127

Scope	Model	MAE min	RMSE min	MAPE %
All test traversals	Gradient Boosting Regressor	3.623	6.320	22.04
All test traversals	Historical-Median Baseline	4.270	6.935	27.49
Route 90: 36 traversals	Gradient Boosting Regressor	1.166	1.599	15.00
Route 90: 36 traversals	Historical-Median Baseline	1.303	2.128	16.08

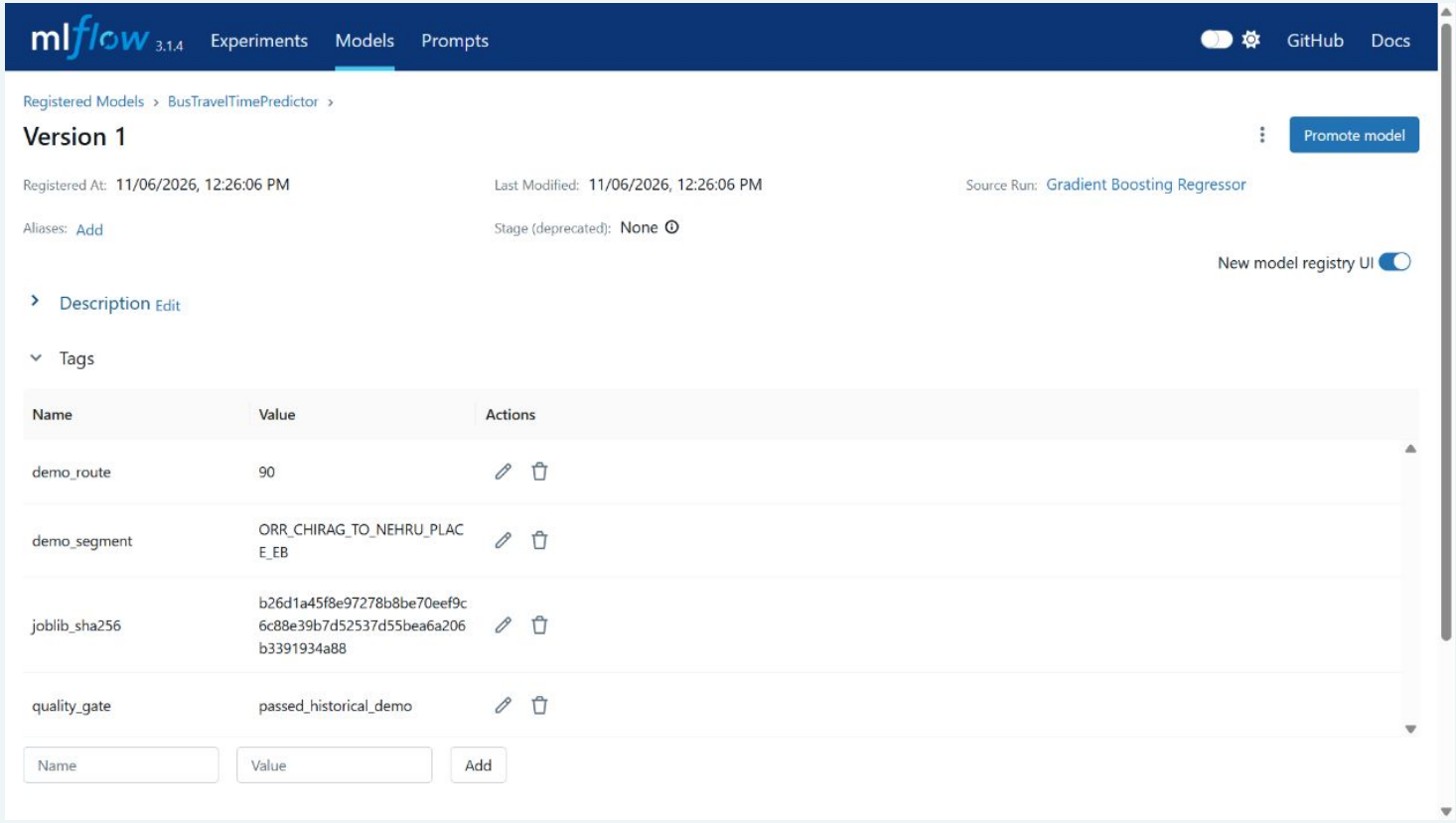
Selection and checks: The candidate's validation MAE was 2.474 minutes. The saved report records 12/12 passing quality checks. Checks include split agreement, feature consistency, candidate improvement, finite positive predictions, reload agreement and logged-artifact hashes. The earlier approval-stage test count is 29; the current suite is recorded on page 1.

Limits: This repeats previously evaluated 2019 dates in the release environment. The demo is scoped to Route 90 eastbound. Other segments remain research coverage. Current service accuracy and timetable-to-gate mapping remain unevaluated. Inference does not consume a live GPS feed. An HTTP success validates serving behavior; historical error metrics support the accuracy claim.

MLflow model registry

Dashboard screenshot from a local registry copy; the original store was checked separately in read-only mode.

Observed: BusTravelTimePredictor v1 - READY; training run 66c819ceb83740c18d05800213524441.
Artifact SHA-256: b26d1a45f8e97278b8be70eef9c6c88e39b7d52537d55bea6a206b3391934a88
Registry / approval identity match: True. Read-only registry check: 07 Nov 2026, 08:53 IST.



FastAPI readiness

The service reports the loaded model version and feature-history policy.

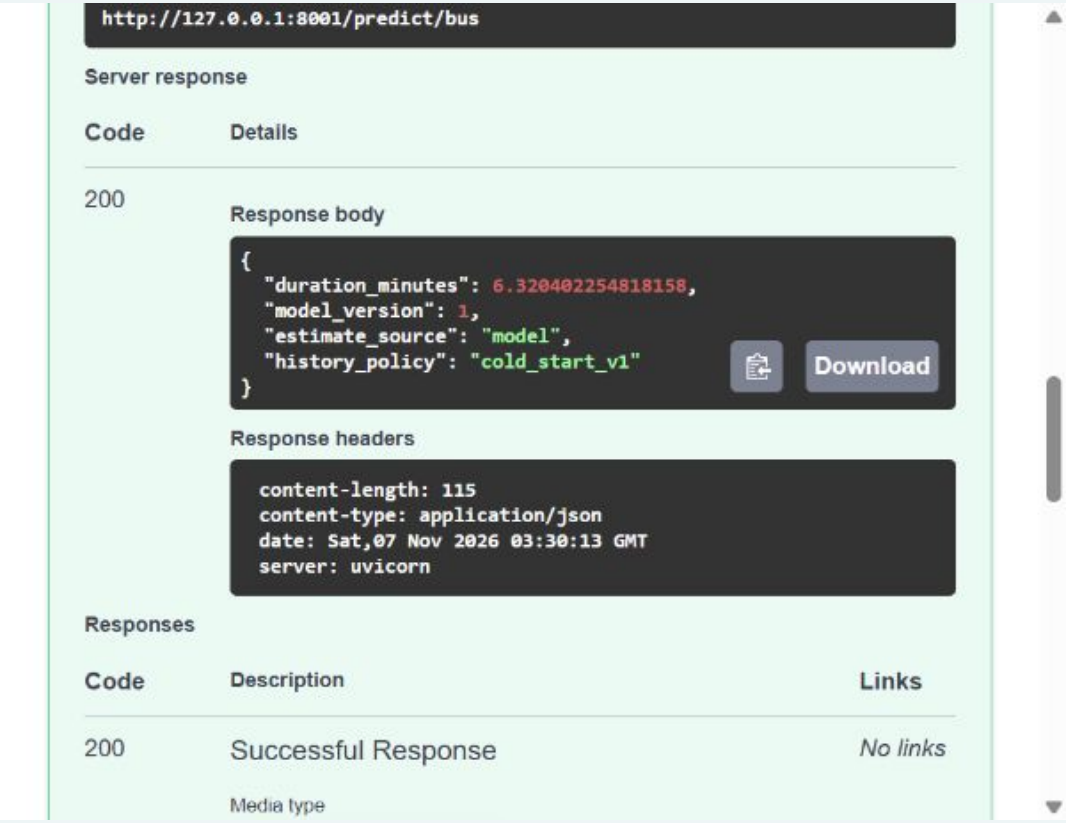
Runtime check: 07 Nov 2026, 08:51 IST.
Recorded GET /health: {"case": "readiness", "status": 200, "response": {"status": "ready", "model_name": "BusTravelTimePredictor", "model_version": 1, "history_policy": "cold_start_v1"}}



Approved bus-time prediction

The published model predicts duration for the documented historical Route 90 request.

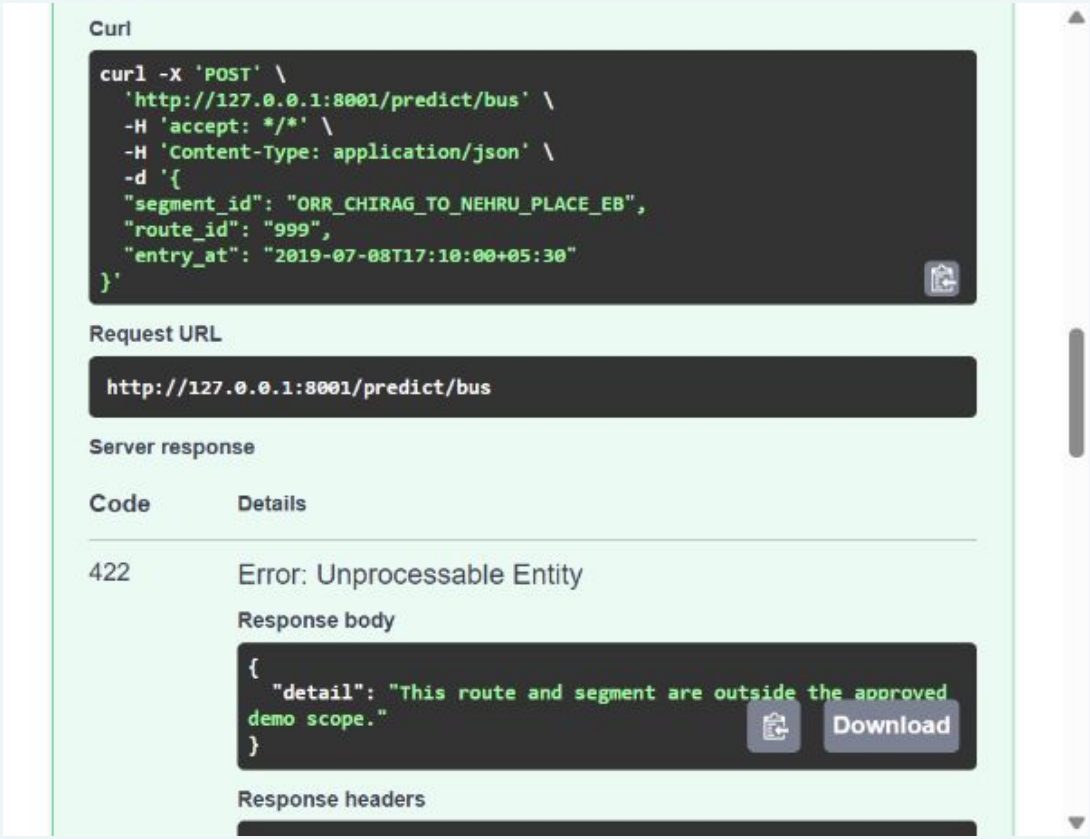
Request: {"segment_id": "ORR_CHIRAG_TO_NEHRU_PLACE_EB", "route_id": "90", "entry_at": "2019-07-08T17:10:00+05:30"}
Observed response (HTTP 200): {"duration_minutes": 6.320402254818158, "model_version": 1, "estimate_source": "model", "history_policy": "cold_start_v1"}



Request validation and model scope

Requests outside the approved route/segment and timestamps without a timezone are rejected.

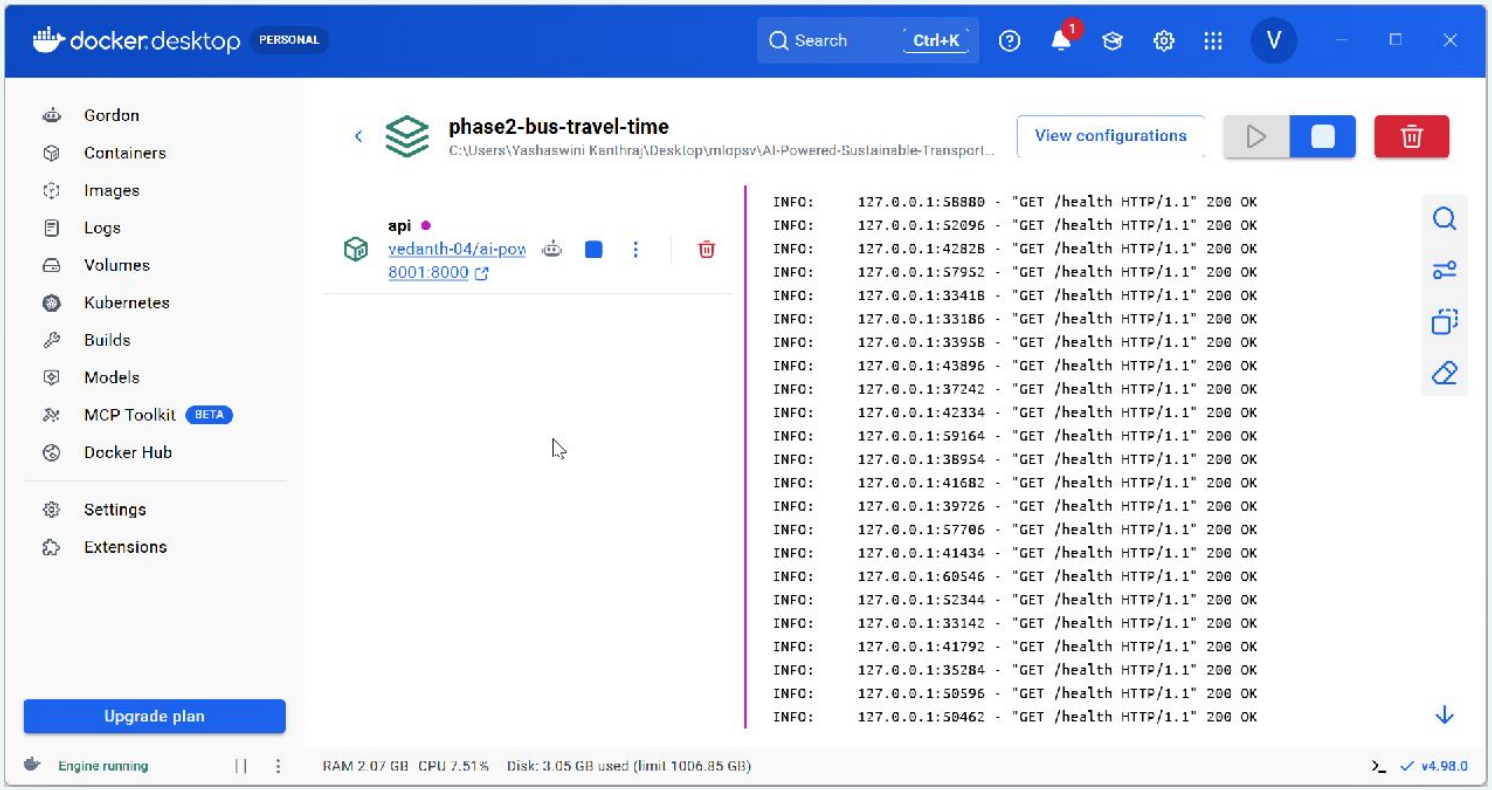
Observed HTTP checks: unsupported_route: HTTP 422; unsupported_segment: HTTP 422; missing_timezone: HTTP 422.
Unsupported-route response: {"detail": "This route and segment are outside the approved demo scope."}



Docker deployment on Windows / WSL 2

The verifier compares the running container and its repository digest with the approved release.

Observed container: running=True, health=healthy; binding {"8000/tcp": [{"HostIp": "127.0.0.1", "HostPort": "8001"}]}.
Approved image:
ghcr.io/vedanth-04/ai-powered-sustainable-transportation-advisor/bus-travel-time@sha256:0250462721176db01d63cd77f93ba7647dd7c94890dc1f7d6e7025e85e2b5dad
Image source revision: 44d31e188ec0e7ad10bf439a4b53f42f0774e63d. Runtime verification overall_passed=True.



Continuous integration

GitHub Actions tests source, builds a fixture image, and checks readiness, prediction and validation over HTTP.

Recorded CI: Phase 2 CI - completed / success; source 33a0fc7da1d42ca44089416500979aaf9a0ac533.
Run: https://github.com/vedanth-04/AI-Powered-Sustainable-Transportation-Advisor/actions/runs/42333748477
The controlled CI fixture expects 6.25 minutes. The real released model has the separate 6.3204-minute sample result shown on page 5.

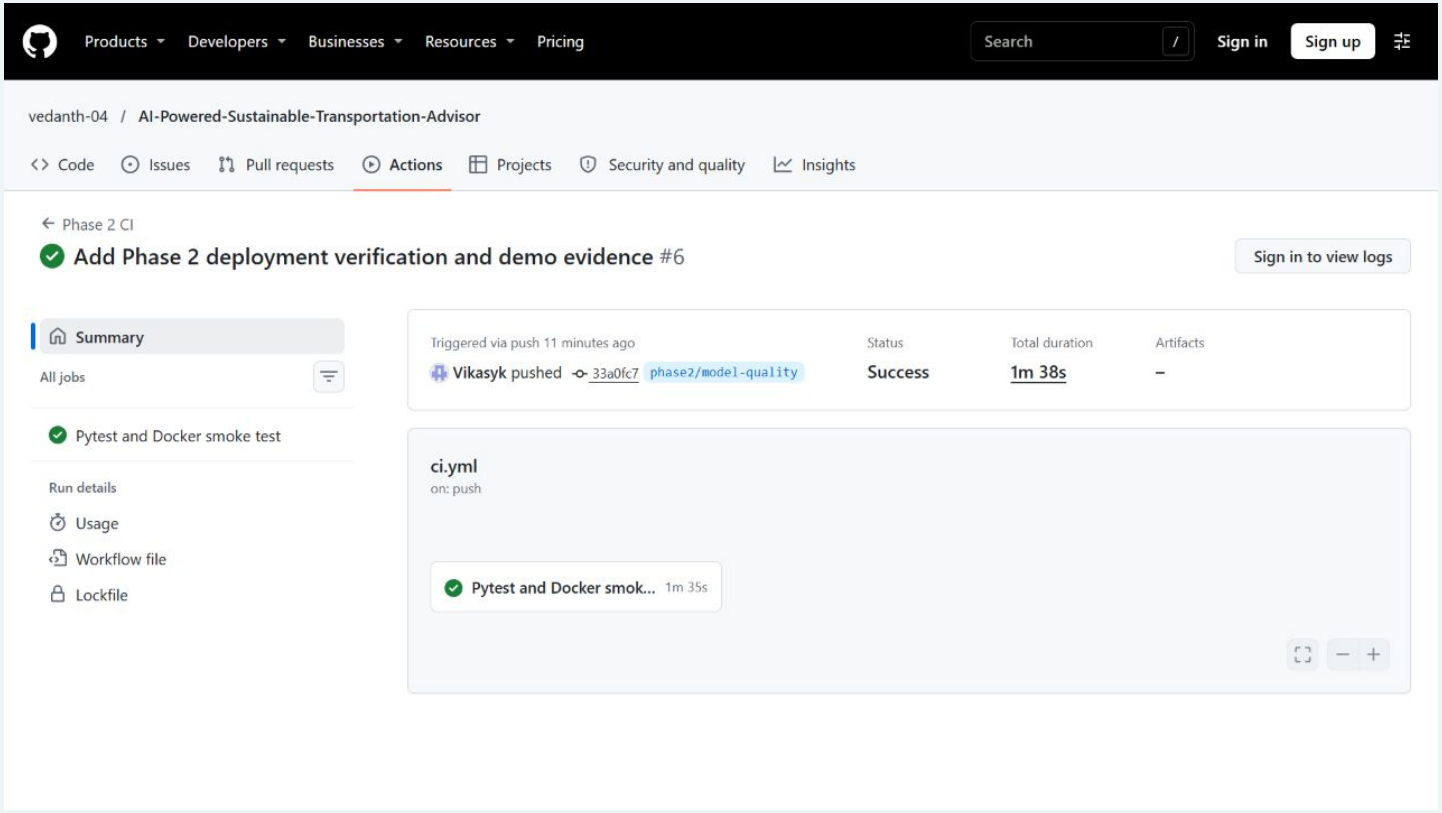


Image release and classroom demonstration

The v1 release publishes the verified real-model image; local deployment then uses its immutable digest.

Release record: https://github.com/vedanth-04/AI-Powered-Sustainable-Transportation-Advisor/actions/runs/42330318360
The release workflow verifies bundle hashes, tests the real model, publishes the image, then pulls and checks the published digest on a fresh runner. Use the visible run result below as publication evidence.
Present in this order: current tests, historical quality, MLflow registry, CI/release, Docker health, valid prediction, HTTP 422. Deployment command: `python -m scripts.deploy_api --port 8001`. Verify with `python -m scripts.verify_deployment`.

